

Discussion 12.1
Application of 2-bit
Flip-Flops

Course Code: CPE 013

Program:

Course Title: Logic Circuits and Design

Date Performed: 1/04/26

Section: 22S2

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Instructions

A. Design a Synchronous Sequential circuit given this states in 2-bits LED: 0 --> 3--> 2 --> 1----> 0.
Apply the different types of flip flops and its excitation table:

(1)JK Flip Flop

(2)T Flip Flop

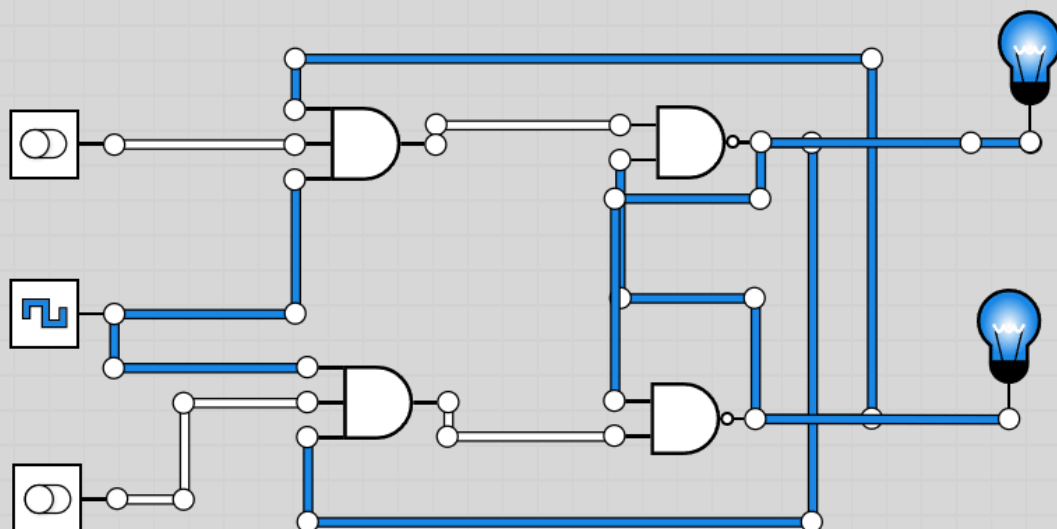
(3) D Flip Flop

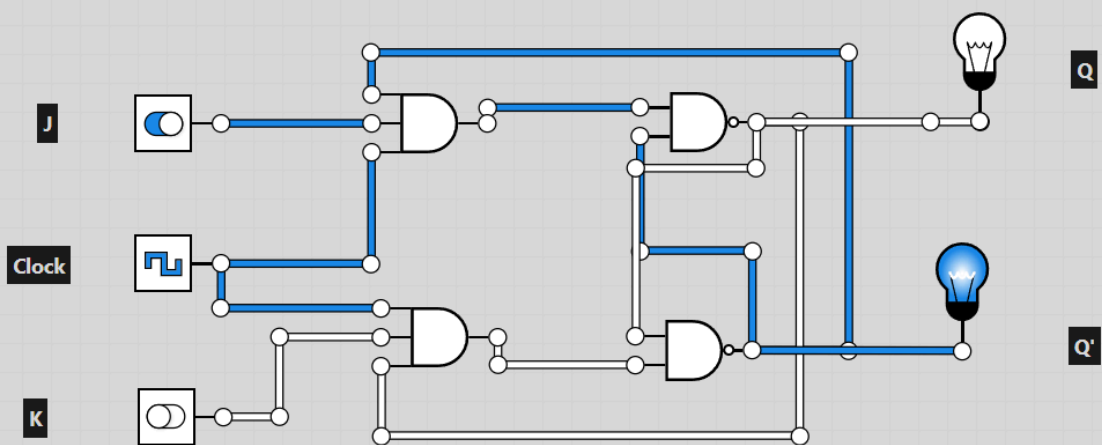
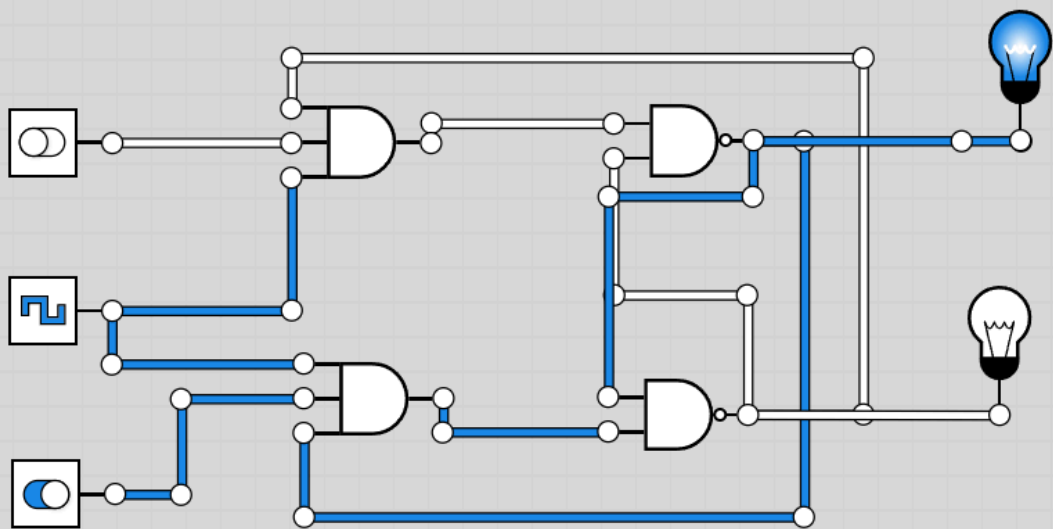
Submit a clear captured image file of the problem, state diagram, state table, kmap, design circuit (HAND-WRITTEN)

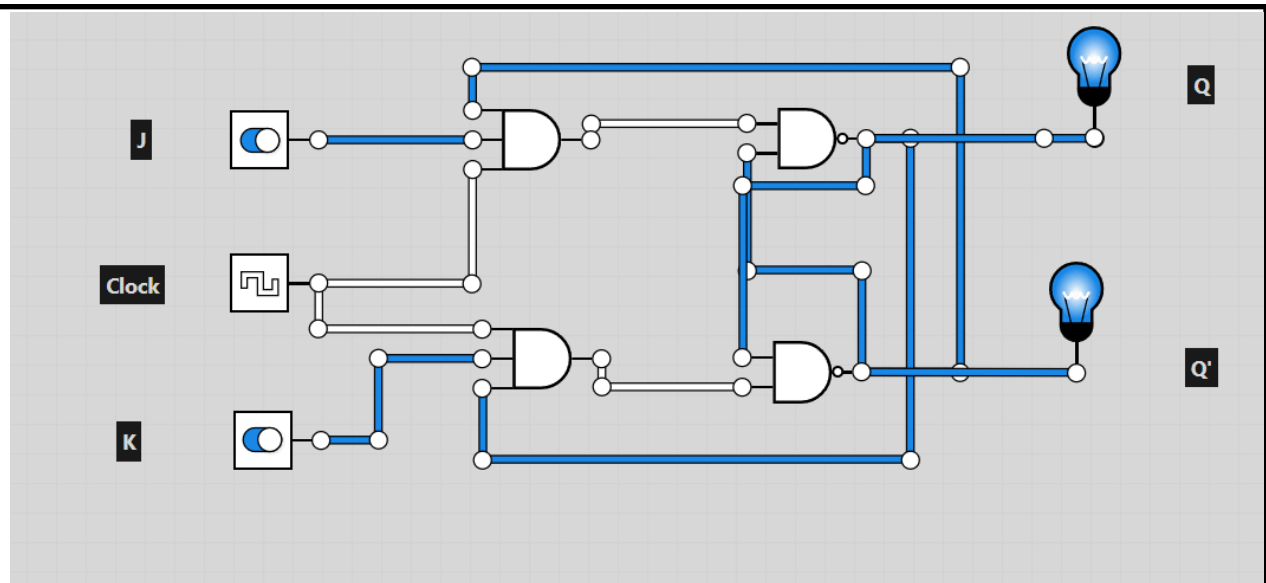
B. Research on the IC and Pin Configurations of these Flip Flops.

Answer

JK Flip-Flop







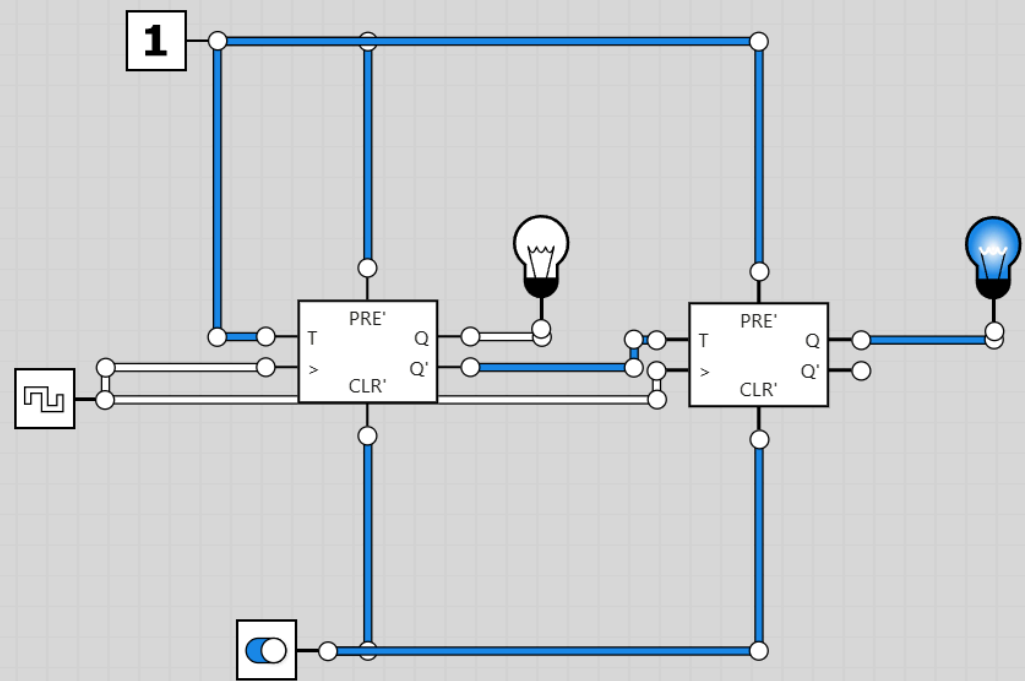
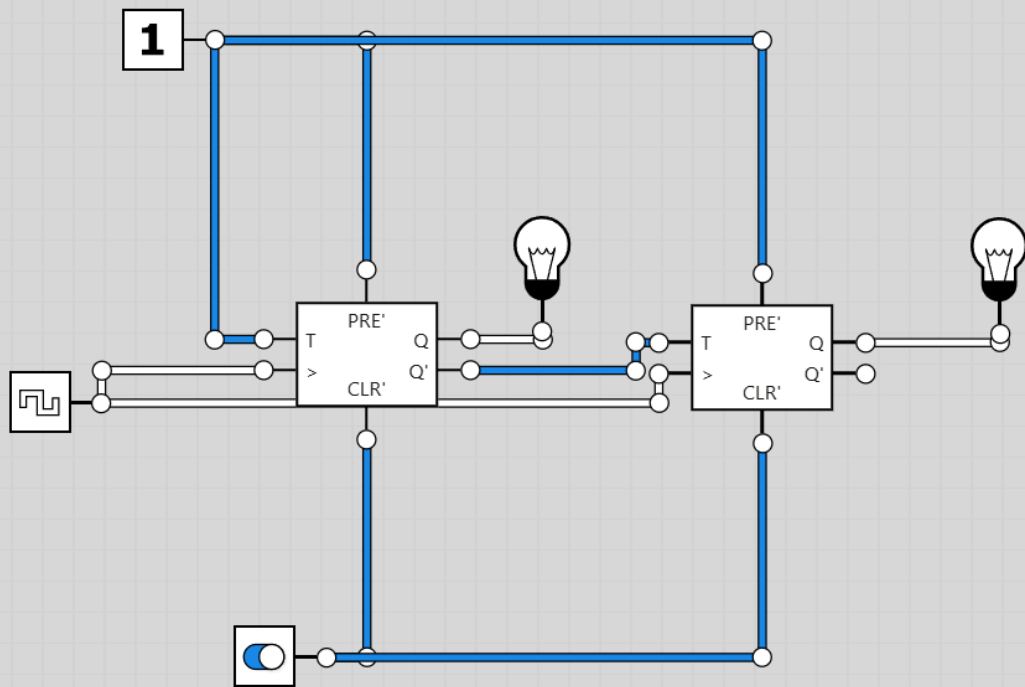
Truth Table

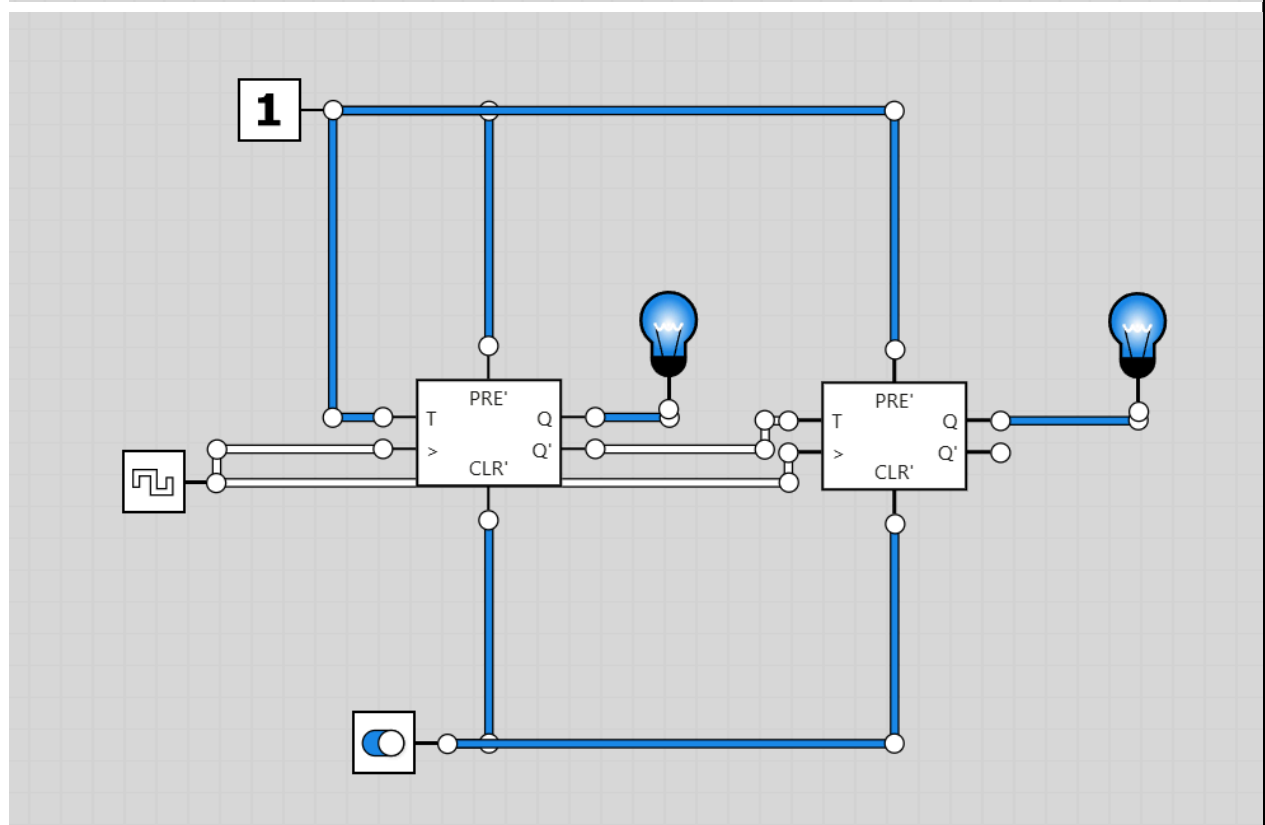
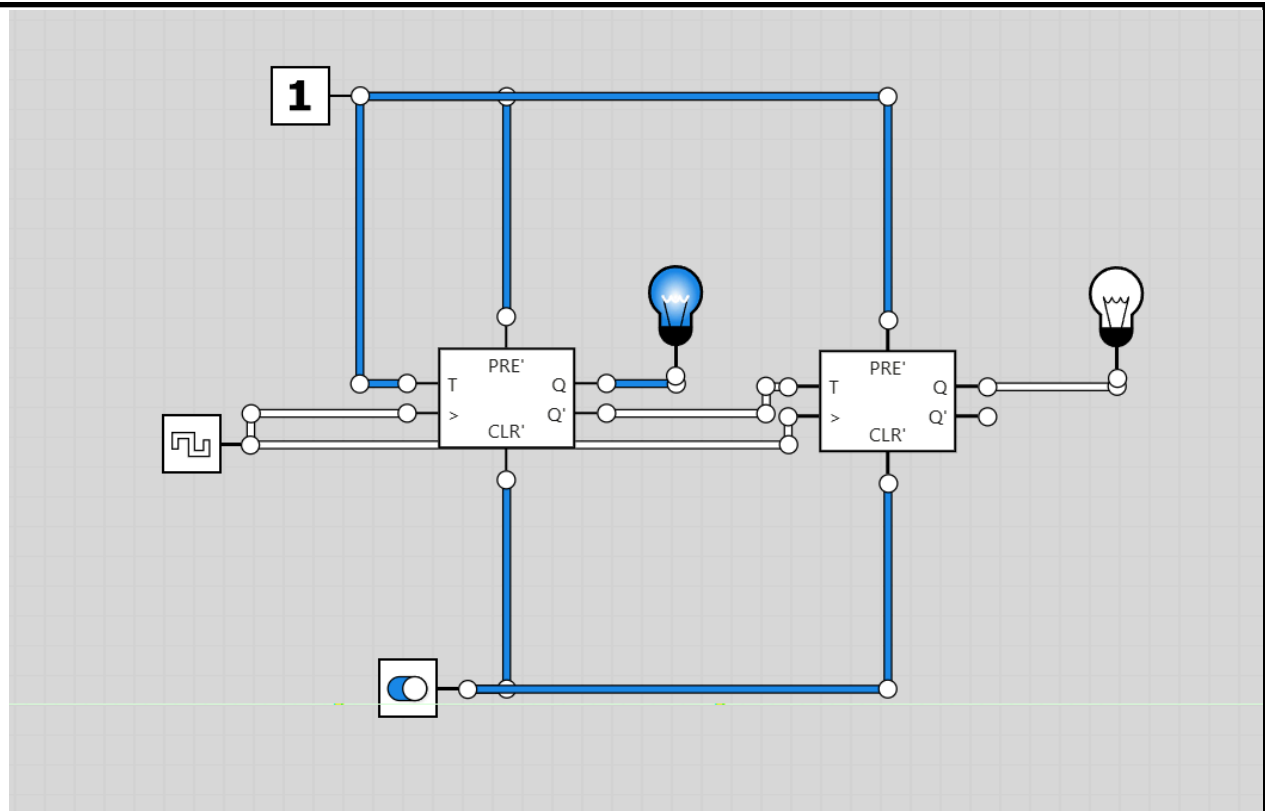
Clock	J	K	Q	Q'
1	0	0	latch	latch
1	0	1	0	1
1	1	0	1	0
1	1	1	Tgl = x	Tgl = x

State Table

Present State	Next State	JA	KA	JB	KB
00	11	1	x	1	x
11	10	x	0	x	1
10	01	x	1	1	x
01	00	0	x	x	1

T Flip-Flop





Truth Table

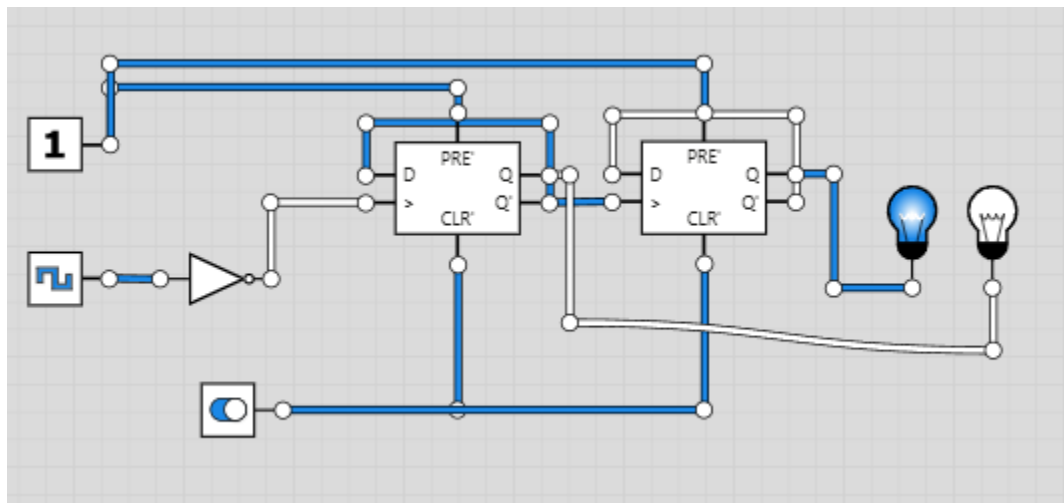
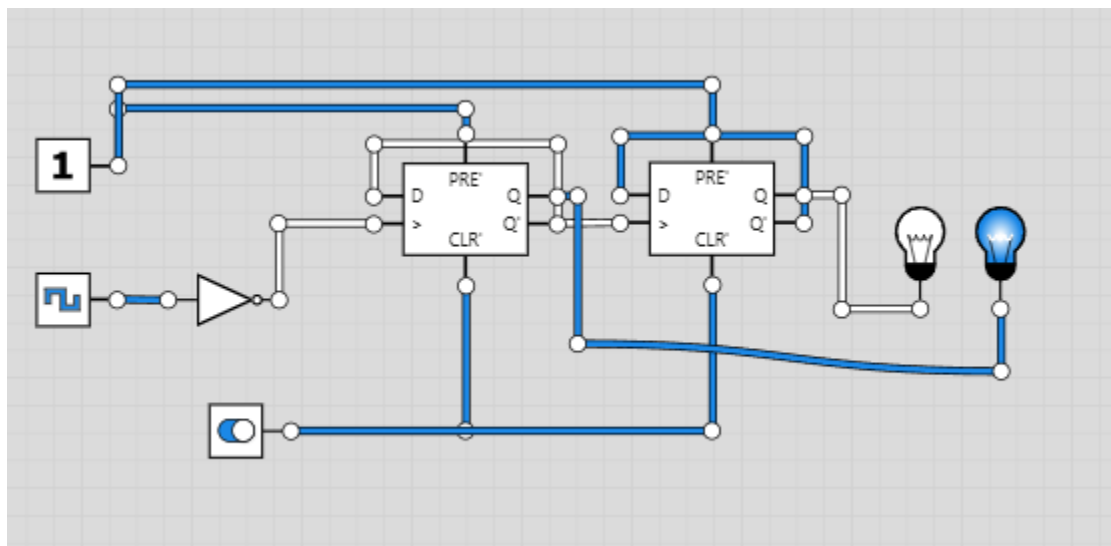
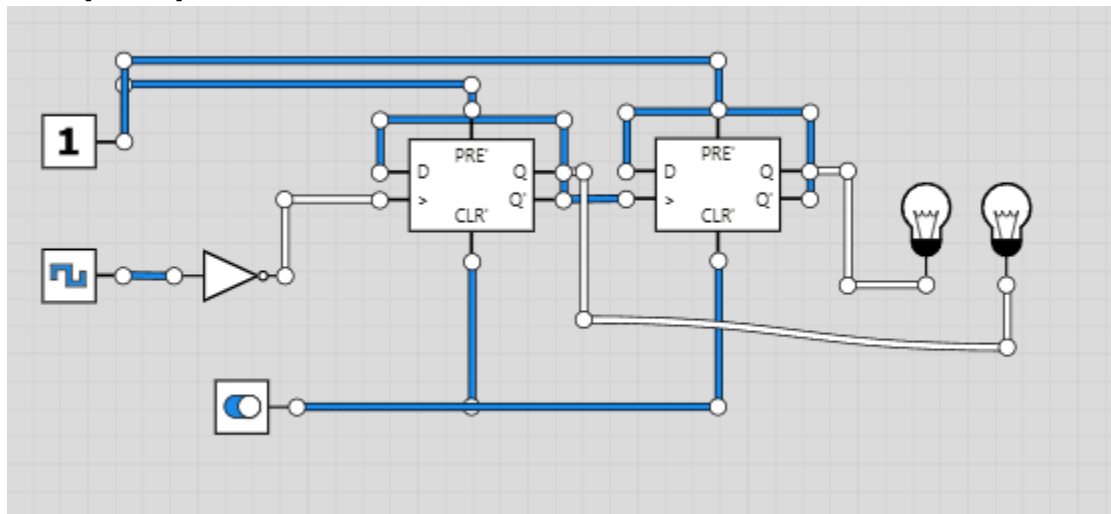
CLK	T	Q(Present State)	Q'(Present State)	Q(Next State)	Q'(Next State)
^	0	0	1	0	1

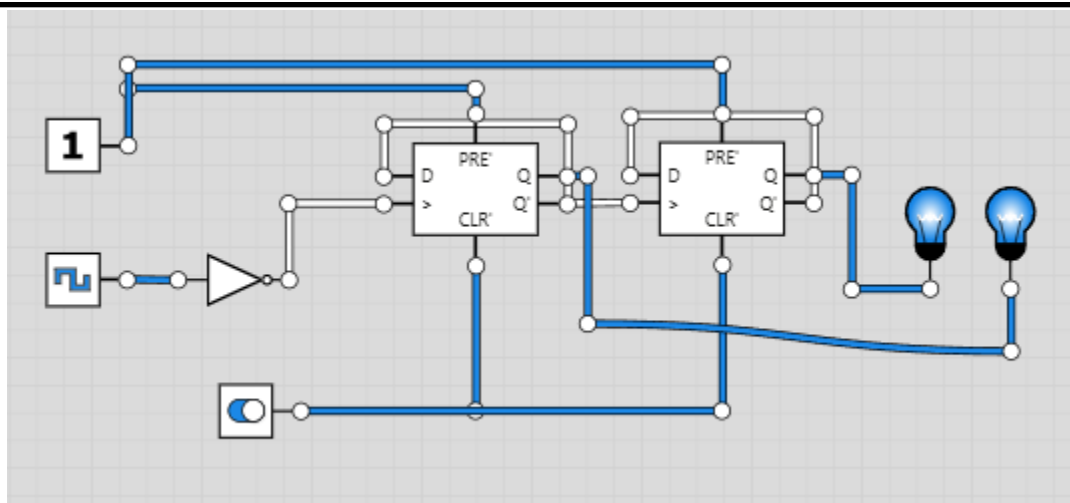
^	0	1	0	1	0
^	1	0	1	1	0
^	1	1	0	0	1

State Table

Present State	Next State	TA	TB
00	11	1	1
11	10	0	1
10	01	1	1
01	00	0	1

D Flip-Flop





Truth Table

CLK	D	Q(Present state)	Q'(Present State)	Q(Next state)	Q'(Next state)
^ 	0	0	1	0	1
^ 	0	1	0	0	1
^ 	1	0	1	1	0
^ 	1	1	0	1	0

State Table

Present State	Next State	DA	DB
00	11	1	1
11	10	1	0
10	01	0	1
01	00	0	0

IC and Pin Configurations of D, T, and J-K Flip-Flops

D Flip-Flop

IC: 7474 (Dual D-Type Positive-Edge-Triggered Flip-Flops with Preset and Clear)

Pin Configs:

- Pin 1: 1CLR' (Clear for FF 1 - Active Low)

- Pin 2: 1D (Data Input for FF 1)
 - Pin 3: 1CLK (Clock Input for FF 1)
 - Pin 4: 1PRE' (Preset for FF 1 - Active Low)
 - Pin 5: 1Q (Output for FF 1)
 - Pin 6: 1Q' (Inverted Output for FF 1)
 - Pin 7: GND (Ground)
 - Pin 8: 2Q' (Inverted Output for FF 2)
 - Pin 9: 2Q (Output for FF 2)
 - Pin 10: 2PRE' (Preset for FF 2 - Active Low)
 - Pin 11: 2CLK (Clock Input for FF 2)
 - Pin 12: 2D (Data Input for FF 2)
 - Pin 13: 2CLR' (Clear for FF 2 - Active Low)
 - Pin 14: VCC (Power Supply, +5V)
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J-K Flip-Flop

IC: 7476 (Dual J-K Flip-Flops with Preset and Clear)

Pin Configs:

- Pin 1: 1CLK (Clock Input for FF 1)
 - Pin 2: 1PRE' (Preset for FF 1 - Active Low)
 - Pin 3: 1CLR' (Clear for FF 1 - Active Low)
 - Pin 4: 1J (J Input for FF 1)
 - Pin 5: VCC (Power Supply, +5V)
 - Pin 6: 2CLK (Clock Input for FF 2)
 - Pin 7: 2PRE' (Preset for FF 2 - Active Low)
 - Pin 8: 2CLR' (Clear for FF 2 - Active Low)
 - Pin 9: 2J (J Input for FF 2)
 - Pin 10: 2Q (Output for FF 2)
 - Pin 11: 2Q' (Inverted Output for FF 2)
 - Pin 12: 2K (K Input for FF 2)
 - Pin 13: GND (Ground)
 - Pin 14: 1Q' (Inverted Output for FF 1)
 - Pin 15: 1Q (Output for FF 1)
 - Pin 16: 1K (K Input for FF 1)
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T Flip-Flop

IC: No dedicated 74-series IC. *T Flip-Flops are almost always created using a 7473 or 7476 (J-K Flip-Flop) by tying the J and K pins together.*

Pin Configs (using 7473 Dual J-K):

- Pin 1: 1CLK (Clock Input for FF 1)
- Pin 2: 1CLR' (Clear for FF 1)
- Pin 3: 1K (Tie to Pin 14 for T input)
- Pin 4: VCC (Power Supply)
- Pin 5: 2CLK (Clock Input for FF 2)
- Pin 6: 2CLR' (Clear for FF 2)
- Pin 7: 2J (Tie to Pin 10 for T input)

- Pin 8: 2Q' (Inverted Output for FF 2)
- Pin 9: 2Q (Output for FF 2)
- Pin 10: 2K (Tie to Pin 7 for T input)
- Pin 11: GND (Ground)
- Pin 12: 1Q (Output for FF 1)
- Pin 13: 1Q' (Inverted Output for FF 1)
- Pin 14: 1J (Tie to Pin 3 for T input)